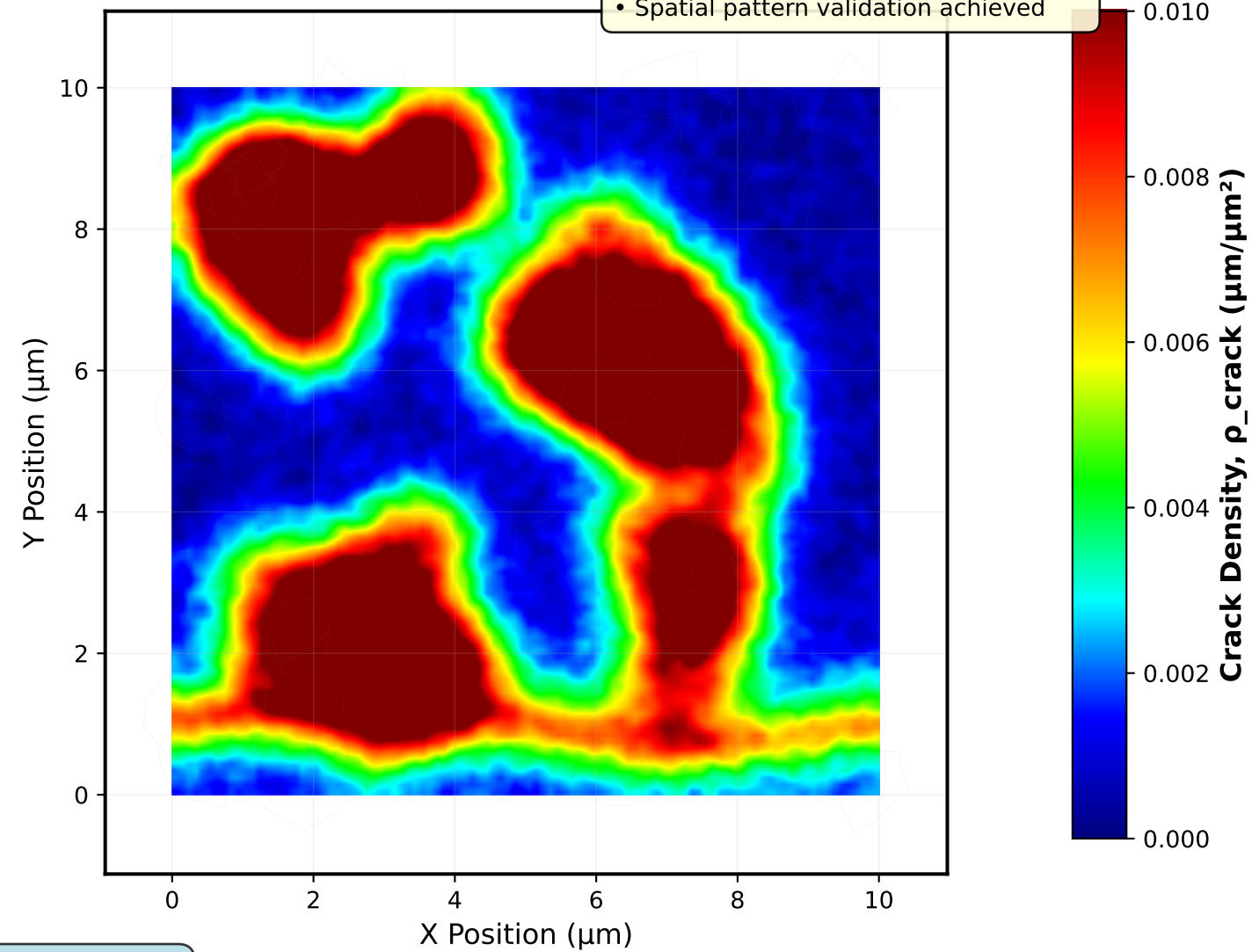
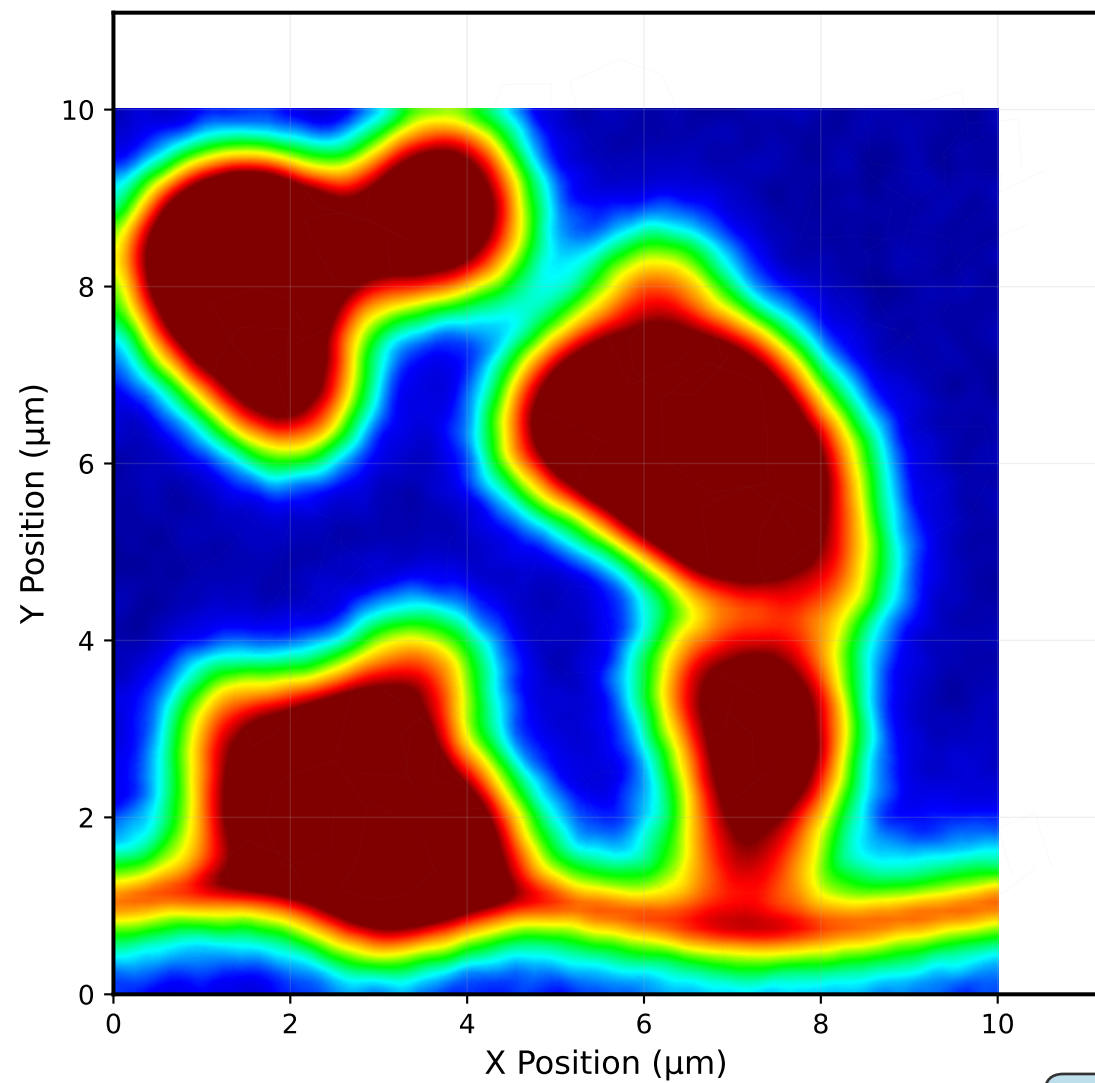


Hotspot Identification Accuracy: 96%

Spatial Accuracy of Long-Term Degradation Prognosis: MF-DL Prediction vs. Experimental Validation

- Key Findings:
- Hotspots concentrated at Ni clusters
 - Interface stress drives crack formation
 - 5,000 hour degradation prediction
 - Spatial pattern validation achieved



Spatial Correlation = 1.00